

Dafa Rafiansyah/1301213290/IF4508

Link google collab :

<https://colab.research.google.com/drive/1OUrEpApX9UVqbwjF0XcqypgP1lp137la?usp=sharing>

The image displays two screenshots of a Google Colab notebook titled "KERJA LAB NAIVE BAYES_1301213290.ipynb".

Top Screenshot: Shows the initial code execution. The code imports pandas, math, and numpy. It defines a function `read_excel` to read Excel files. Then, it loads data from "newTrain.xlsx", "newTest.xlsx", and "newTestGroundTruth.xlsx". The output shows the first five rows of the training data:

Unnamed: 0	id	x1	x2	x3	y
0	0	1	83	57	91
1	1	2	14	15	88
2	2	3	41	12	37
3	3	4	16	29	32
4	4	5	88	21	18

Bottom Screenshot: Shows the next step in the notebook. It displays the first five rows of the test data and the dimensions of the dataText variable (296 rows x 6 columns). The output shows the first ten rows of the test data:

Unnamed: 0	id	x1	x2	x3	y
0	0	297	57	85	2
1	1	298	54	66	73
2	2	299	44	31	15
3	3	300	84	23	17
4	4	301	100	81	55
5	5	302	32	52	45
6	6	303	99	15	26
7	7	304	64	81	40
8	8	305	93	9	80
9	9	306	24	67	53

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RAM Disk

Files

sample_data

Naive-Bayes-Classifica...

newTest.xlsx

newTestGroundTruth...

newTrain.xlsx

[7] dataTruth

Unnamed: 0 id x1 x2 x3 y

0 0 297 82 73 78 1

1 1 298 31 38 68 1

2 2 299 64 91 73 1

3 3 300 33 35 26 0

4 4 301 11 82 30 0

5 5 302 88 26 36 1

6 6 303 22 46 46 1

7 7 304 14 54 12 1

8 8 305 34 54 67 0

9 9 306 88 96 60 0

[8] #melakukan data eksplorasi

dataTrain.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 296 entries, 0 to 295

Data columns (total 6 columns):

column Non-Null Count Dtype

0 Unnamed: 0 296 non-null int64

1 id 296 non-null int64

2 x1 296 non-null int64

3 x2 296 non-null int64

4 x3 296 non-null int64

5 y 296 non-null int64

dtypes: int64(6)

memory usage: 14.0 KB

[11] dataText.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 6 columns):

Column Non-Null Count Dtype

0 Unnamed: 0 10 non-null int64

1 id 10 non-null int64

2 x1 10 non-null int64

3 x2 10 non-null int64

4 x3 10 non-null int64

5 y 10 non-null object

dtypes: int64(5), object(1)

memory usage: 608.0+ bytes

[12] dataTruth.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 6 columns):

Column Non-Null Count Dtype

0 Unnamed: 0 10 non-null int64

1 id 10 non-null int64

2 x1 10 non-null int64

3 x2 10 non-null int64

4 x3 10 non-null int64

5 y 10 non-null object

dtypes: int64(5), object(1)

memory usage: 608.0+ bytes

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21:37 27/05/2023

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RAM Disk

Files

sample_data

Naive-Bayes-Classifica...

newTest.xlsx

newTestGroundTruth...

newTrain.xlsx

[8] <class 'pandas.core.frame.DataFrame'>

RangeIndex: 296 entries, 0 to 295

Data columns (total 6 columns):

column Non-Null Count Dtype

0 Unnamed: 0 296 non-null int64

1 id 296 non-null int64

2 x1 296 non-null int64

3 x2 296 non-null int64

4 x3 296 non-null int64

5 y 296 non-null int64

dtypes: int64(6)

memory usage: 14.0 KB

[11] dataText.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 6 columns):

Column Non-Null Count Dtype

0 Unnamed: 0 10 non-null int64

1 id 10 non-null int64

2 x1 10 non-null int64

3 x2 10 non-null int64

4 x3 10 non-null int64

5 y 10 non-null object

dtypes: int64(5), object(1)

memory usage: 608.0+ bytes

[12] dataTruth.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 6 columns):

Column Non-Null Count Dtype

0 Unnamed: 0 10 non-null int64

1 id 10 non-null int64

2 x1 10 non-null int64

3 x2 10 non-null int64

4 x3 10 non-null int64

5 y 10 non-null object

dtypes: int64(5), object(1)

memory usage: 608.0+ bytes

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21:37 27/05/2023

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File Edit View Insert Runtime Tools Help All changes saved

Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
memory usage: 608.0+ bytes
```

```
[12] dataTruth.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   Unnamed: 0  10 non-null    int64
 1   id          10 non-null    int64
 2   x1         10 non-null    int64
 3   x2         10 non-null    int64
 4   x3         10 non-null    int64
 5   y          10 non-null    int64
dtypes: int64(6)
memory usage: 608.0 bytes
```

Data Preprocessing

```
[13] #drop kolom dataset yg tdk diperlukan
dataTrain = dataTrain.drop(["Unnamed: 0"],axis = 1)
dataText = dataText.drop(["Unnamed: 0"], axis = 1)
dataTruth = dataTruth.drop(["Unnamed: 0"], axis = 1)
```

```
[14] dataTrain.head()
```

	id	x1	x2	x3	y
0	1	83	57	81	1

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
[14] dataTrain.head()
```

	id	x1	x2	x3	y
0	1	83	57	81	1
1	2	14	15	88	0
2	3	41	12	37	1
3	4	16	29	32	0
4	5	88	21	18	1

```
[15] dataText.head()
```

	id	x1	x2	x3	y
0	297	57	85	2	?
1	298	54	66	73	?
2	299	44	31	15	?
3	300	84	23	17	?
4	301	100	81	55	?

```
[16] dataTruth.head()
```

	id	x1	x2	x3	y
0	297	57	85	2	?
1	298	54	66	73	?
2	299	44	31	15	?
3	300	84	23	17	?
4	301	100	81	55	?

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Files

- sample_data
- Naive-Bayes-Classifica...
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- newTestGroundTruth...
- newTrain.xlsx

Code

```
dataTrain.head()
```

	id	x1	x2	x3	y
0	297	82	73	78	1
1	298	31	38	68	1
2	299	64	91	73	1
3	300	33	35	26	0
4	301	11	82	30	0

```
[17] def normalizationMinMax(allData, columnTarget):  
    for column in columnTarget:  
        allData[column] = (  
            (allData[column] - allData[column].min())  
            / allData[column].max() - allData[column].min()  
        )  
    return allData  
  
[18] #melakukan combine untuk melakukan normalisasi  
testStartID = dataTrain.index.stop  
  
allData = pd.concat([dataTrain, dataTest])  
  
allData = normalizationMinMax(allData, ['x1', 'x2', 'x3'])  
  
normTrain, normTest = allData.iloc[:testStartID].drop('id', axis=1), allData.iloc[testStartID:].drop('id', axis=1)
```

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21:37 27/05/2023

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
normTrain
```

	x1	x2	x3	y
0	0.83	57	91	1
1	0.14	15	88	0
2	0.41	12	37	1
3	0.16	29	32	0
4	0.88	21	18	1
...	...	...	...	...
291	0.29	16	27	1
292	0.19	2	93	1
293	0.68	34	67	0
294	0.50	97	15	1
295	0.90	16	16	1
296 rows x 4 columns				

```
[20] normTest
```

	x1	x2	x3	y
0	0.57	85	2	?

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21:38 27/05/2023

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[20] normTest

	x1	x2	x3	y
0	0.57	85	2	?
1	0.54	66	73	?
2	0.44	31	15	?
3	0.84	23	17	?
4	1.00	81	55	?
5	0.32	52	45	?
6	0.99	15	26	?
7	0.64	81	40	?
8	0.93	9	80	?
9	0.24	67	53	?

[21] def standardization(allData, columnTarget):
for column in columnTarget:
allData[column] = (allData[column]-allData[column].mean())/(allData[column].std())
return allData

[22] testStartID = dataTrain.index.stop
allData = pd.concat([dataTrain, dataTest])

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RAM Disk

Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[22] allData = standardization(allData,['x1','x2','x3'])
stdTrain, stdTest = allData.iloc[:testStartID].drop('id',axis=1),allData.iloc[testStartID:].drop('id',axis=1)

stdTrain

	x1	x2	x3	y
0	1.164512	0.226265	1.440004	1
1	-1.264364	-1.189489	1.337972	0
2	-0.313934	-1.290615	-0.396568	1
3	-1.193961	-0.717571	-0.566621	0
4	1.340518	-0.987239	-1.042769	1
...	...	...	...	...
291	-0.736347	-1.155781	-0.736674	1
292	-1.088358	-1.627699	1.508025	1
293	0.636496	-0.549029	0.623750	0
294	0.002876	1.574603	-1.144801	1
295	1.410920	-1.155781	-1.110790	1
...	...	...	...	...
296	...	...	...	...

296 rows x 4 columns

[24] stdTest

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[24] stdTest

	x1	x2	x3	y
0	0.249284	1.170102	-1.586938	?
1	0.143680	0.529641	0.827813	?
2	-0.208331	-0.650154	-1.144801	?
3	1.199713	-0.919822	-1.076780	?
4	1.762931	1.035268	0.215623	?
5	-0.630744	0.057723	-0.124483	?
6	1.727730	-1.189489	-0.770684	?
7	0.495691	1.035268	-0.294536	?
8	1.516523	-1.391740	1.065887	?
9	-0.912353	0.563350	0.147601	?

```
[26] def splitTruth(dataTrain, columnTarget):
      truthData = []
      for truth in dataTrain[columnTarget].unique():
          truthData.append(dataTrain.where(dataTrain[columnTarget]==truth).dropna())
      return truthData

      yesData, noData = splitTruth(dataTrain, columnTarget='y')
```

[27] yesData

84.41 GB available

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[27] yesData

	id	x1	x2	x3	y
0	1.0	83.0	57.0	91.0	1.0
2	3.0	41.0	12.0	37.0	1.0
4	5.0	88.0	21.0	18.0	1.0
5	6.0	3.0	18.0	51.0	1.0
7	8.0	11.0	7.0	1.0	1.0
...	...	...	...	...	...
290	291.0	6.0	20.0	87.0	1.0
291	292.0	29.0	16.0	27.0	1.0
292	293.0	19.0	2.0	93.0	1.0
294	295.0	50.0	97.0	15.0	1.0
295	296.0	90.0	16.0	16.0	1.0

151 rows x 5 columns

[28] noData

	id	x1	x2	x3	y
1	2.0	14.0	15.0	88.0	0.0
3	4.0	16.0	29.0	32.0	0.0

84.41 GB available

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RAM Disk

Files

sample_data

Naive-Bayes-Classifica...

newTest.xlsx

newTestGroundTruth...

newTrain.xlsx

Code

Text

28

1d x1 x2 x3 y

1 2.0 14.0 15.0 88.0 0.0

3 4.0 16.0 29.0 32.0 0.0

6 7.0 29.0 100.0 14.0 0.0

10 11.0 22.0 36.0 69.0 0.0

14 15.0 40.0 51.0 21.0 0.0

...

283 284.0 48.0 99.0 1.0 0.0

284 285.0 13.0 0.0 23.0 0.0

287 288.0 86.0 48.0 54.0 0.0

288 289.0 83.0 40.0 73.0 0.0

293 294.0 68.0 34.0 67.0 0.0

145 rows x 5 columns

34

def find_mean(yesData, noData, columnTarget):

yesMean = dict()

noMean = dict()

for column in columnTarget:

yesMean[column] = yesData[column].mean()

noMean[column] = noData[column].mean()

return yesMean, noMean

yesMean, noMean = find_mean(yesData, noData, columnTarget=['x1', 'x2', 'x3'])

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21:38 27/05/2023

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KERJA LAB NAIVE BAYES_1301213290.ipynb

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Comment Share

RAM Disk

Files

sample_data

Naive-Bayes-Classifica...

newTest.xlsx

newTestGroundTruth...

newTrain.xlsx

Code

Text

35

print(f'Mean Result\n1 : {yesMean}\n0 : {noMean}')

Mean Result

1 : {'x1': 47.95364238410596, 'x2': 48.741721854304636, 'x3': 51.841059602649004}

0 : {'x1': 58.91724137931035, 'x2': 51.84827586206897, 'x3': 45.90344827586207}

36

def find_std(yesData, noData, columnTarget):

yesStd = dict()

noStd = dict()

for column in columnTarget:

yesStd[column] = yesData[column].std()

noStd[column] = noData[column].std()

return yesStd, noStd

yesStd, noStd = find_std(yesData, noData, columnTarget=['x1', 'x2', 'x3'])

37

print(f'Standard Deviation Result\n1 : {yesStd}\n0 : {noStd}')

Standard Deviation Result

1 : {'x1': 28.886753076648443, 'x2': 29.63386881619497, 'x3': 29.41475654955786}

0 : {'x1': 27.766512466627617, 'x2': 29.84900217820493, 'x3': 29.42913431210903}

38

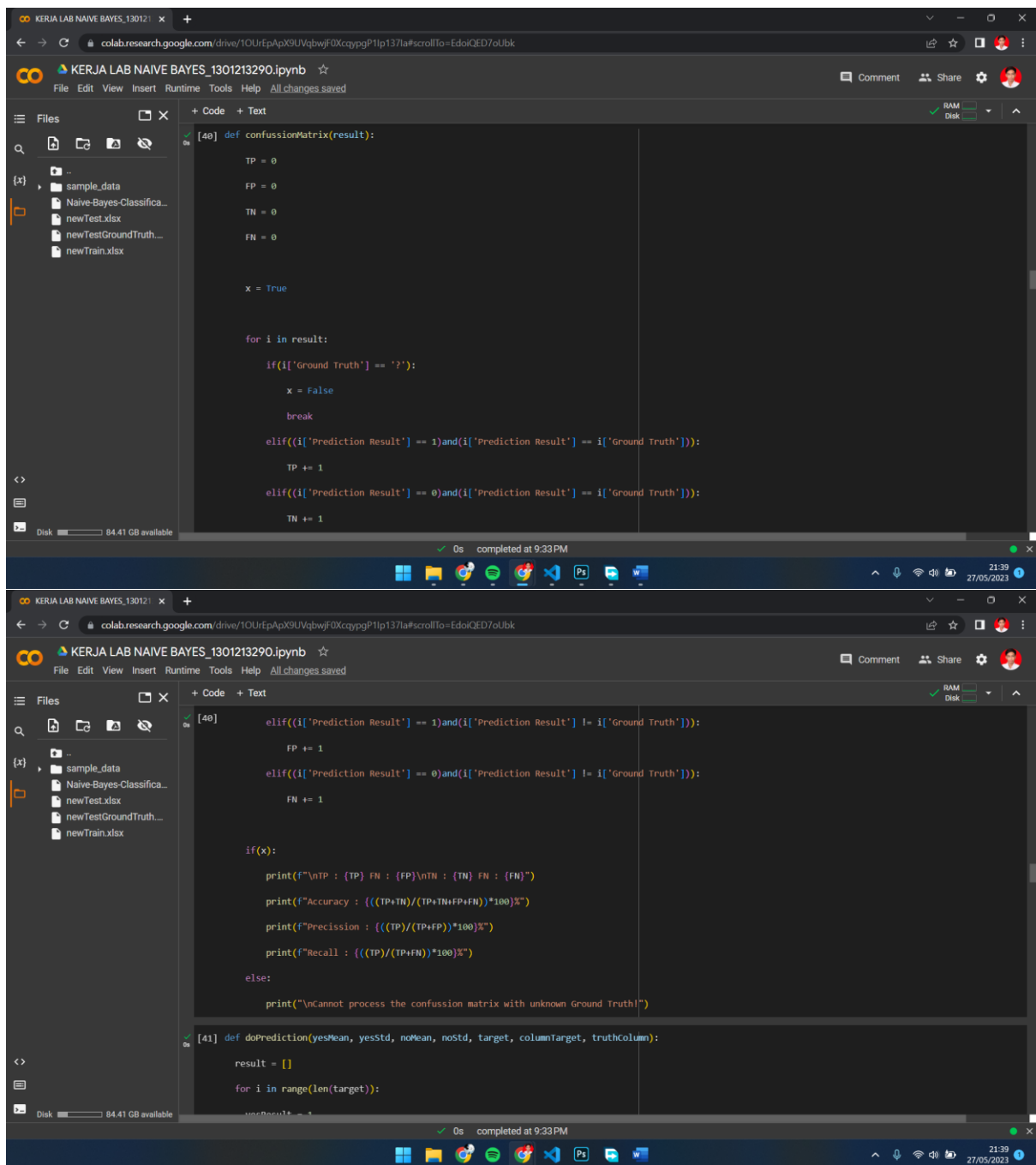
def calc_probality(mean, std, x):

exponent = math.exp(-(x-mean)**2/(2*std**2))

return (1/(math.sqrt(2*math.pi)*std))*exponent

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21:39 27/05/2023



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KERJA LAB NAIVE BAYES_1301213290.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Files

sample_data

Naive-Bayes-Classifica...

newTest.xlsx

newTestGroundTruth...

newTrain.xlsx

+ Code + Text

```
[41] yesResult = 1
      noResult = 1

      for column in columnTarget:

          yesResult *= calc_probability(yesMean[column], yesStd[column], target[column].iloc[i])
          noResult *= calc_probability(noMean[column], noStd[column], target[column].iloc[i])

          result.append({'ID' : target['id'].iloc[i], 'Yes Probability' : "{}".format(yesResult), 'No Probability' : "{}".format(noResult),
                        'Prediction Result' : int(yesResult > noResult),
                        'Ground Truth' : target[truthColumn].iloc[i]})

      return result

[43] # result
      result = []
      target = dataText

      result = doPrediction(yesMean, yesStd, noMean, noStd, target, columnTarget=['x1','x2','x3'], truthColumn='y')

      for p in result:
```

0s completed at 9:33 PM

21:39 27/05/2023

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KERJA LAB NAIVE BAYES_1301213290.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Files

sample_data

Naive-Bayes-Classifica...

newTest.xlsx

newTestGroundTruth...

newTrain.xlsx

+ Code + Text

```
print(p)

confusionMatrix(result)

{'ID': 297, 'Yes Probability': '2.703055728386299e-07', 'No Probability': '4.5075808544073367e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 298, 'Yes Probability': '1.607512382103284e-06', 'No Probability': '1.513301932585517e-06', 'Prediction Result': 1, 'Ground Truth': '?'}
{'ID': 299, 'Yes Probability': '9.531080657381655e-07', 'No Probability': '1.1393103292410503e-06', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 300, 'Yes Probability': '3.9350076167602443e-07', 'No Probability': '4.95396790081609e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 301, 'Yes Probability': '2.73492366379816e-07', 'No Probability': '3.2292632787827514e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 302, 'Yes Probability': '2.094468858775563e-06', 'No Probability': '2.062990263825832e-06', 'Prediction Result': 1, 'Ground Truth': '?'}
{'ID': 303, 'Yes Probability': '1.8813904672513574e-07', 'No Probability': '2.1581843864443093e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 304, 'Yes Probability': '1.1019747916698164e-06', 'No Probability': '1.4172159083752534e-06', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 305, 'Yes Probability': '1.9234460333525067e-07', 'No Probability': '1.50575323199431e-07', 'Prediction Result': 1, 'Ground Truth': '?'}
{'ID': 306, 'Yes Probability': '1.4777286301728196e-06', 'No Probability': '1.3894897355371137e-06', 'Prediction Result': 1, 'Ground Truth': '?'}

Cannot process the confusion matrix with unknown Ground Truth!

[44] def folding(dataset, trainingPercentage, location, shuffle:bool):

      lengthTraining = int(len(dataset)*trainingPercentage/100)

      # randomize the the data position

      if(shuffle):

          dataset = dataset.sample(frac=1).reset_index(drop=True)

      train = []

      validation = []
```

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21:39 27/05/2023

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KERJA LAB NAIVE BAYES_1301213290.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Comment Share

Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

```
validation = []
if(location == 'left'):
    train, validation = dataset.iloc[:lengthTraining].reset_index(drop=True), dataset.iloc[lengthTraining:].reset_index(drop=True)
elif(location == 'right'):
    validation, train = dataset.iloc[abs(lengthTraining-len(dataset))].reset_index(drop=True), dataset.iloc[abs(lengthTraining-len(dataset))].reset_i
elif(location == 'middle'):
    train = dataset.iloc[int(abs(lengthTraining-len(dataset))/2):len(dataset)-int(abs(lengthTraining-len(dataset))/2)]
    validation = pd.concat([dataset.iloc[:int(abs(lengthTraining-len(dataset))/2)], dataset.iloc[len(dataset)-int(abs(lengthTraining-len(dataset))/2):]]
return train, validation

[45] trainData, validationData = folding(dataTrain.copy(), 70, 'left', shuffle=True)

[46] trainData
```

	id	x1	x2	x3	y
0	274	13	79	87	1
1	185	30	84	45	1
2	248	21	65	40	1
3	150	19	70	6	1
4	271	83	58	56	1

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Files

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- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

```
[46] trainData
```

	id	x1	x2	x3	y
4	271	83	58	56	1
...	...	...	...	...	...
202	72	73	14	45	1
203	238	55	33	52	0
204	65	13	47	18	1
205	86	27	61	9	1
206	154	70	96	86	1

207 rows x 5 columns

```
[47] validationData
```

	id	x1	x2	x3	y
0	133	16	72	68	1
1	220	43	23	66	1
2	218	95	54	54	1
3	132	63	81	5	0
4	162	97	35	51	0
...	...	...	...	...	...
84	102	31	58	42	0
85	6	3	18	51	1

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- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
[47] 86 275 64 42 38 0
      87 227 98 81 8 0
      88 174 46 89 86 0
      89 rows x 5 columns
```

```
[48] yesDataTrain, noDataTrain = splitTruth(trainData, columnTarget='y')
      yesDataTrain
```

	id	x1	x2	x3	y
0	274.0	13.0	79.0	87.0	1.0
1	185.0	30.0	84.0	45.0	1.0
2	248.0	21.0	65.0	40.0	1.0
3	150.0	19.0	70.0	6.0	1.0
4	271.0	83.0	58.0	56.0	1.0
...	...	...	...	...	...
199	42.0	63.0	21.0	52.0	1.0
202	72.0	73.0	14.0	45.0	1.0
204	65.0	13.0	47.0	18.0	1.0
205	86.0	27.0	61.0	9.0	1.0
206	154.0	70.0	96.0	86.0	1.0

111 rows x 5 columns

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
noDataTrain
```

	id	x1	x2	x3	y
8	27.0	24.0	75.0	66.0	0.0
9	155.0	18.0	33.0	15.0	0.0
10	172.0	21.0	1.0	25.0	0.0
13	15.0	40.0	51.0	21.0	0.0
18	243.0	13.0	16.0	17.0	0.0
...	...	...	...	...	...
197	115.0	60.0	92.0	66.0	0.0
198	21.0	12.0	17.0	38.0	0.0
200	50.0	8.0	93.0	90.0	0.0
201	33.0	16.0	58.0	94.0	0.0
203	238.0	55.0	33.0	52.0	0.0

96 rows x 5 columns

```
[50] # Split yes and no
      yesDataTrain, noDataTrain = splitTruth(trainData, columnTarget='y')
```

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
# Split yes and no
yesDataTrain, noDataTrain = splitTruth(trainData, columnTarget='y')

# Find mean
yesMeanTrain, noMeanTrain = find_mean(yesDataTrain, noDataTrain, columnTarget=['x1','x2','x3'])

# Find standard deviation
yesStdTrain, noStdTrain = find_std(yesDataTrain, noDataTrain, ['x1','x2','x3'])

result = []

target = validationData # your ground truth data

result = doPrediction(yesMeanTrain, yesStdTrain, noMeanTrain, noStdTrain, target, columnTarget=['x1','x2','x3'], truthColumn='y')

for p in result:

    print(p)

confusionMatrix(result)
```

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
confusionMatrix(result)
```

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ID	Yes Probability	No Probability	Prediction Result	Ground Truth
294	'1.4928181215293987e-06'	'1.428294729092959e-06'	1	0
246	'1.1346911895614773e-06'	'9.167404792591337e-07'	1	1
37	'2.374377501653588e-06'	'2.3766624374103905e-06'	0	0
254	'1.613913439596947e-06'	'1.4810429171965808e-06'	1	0
91	'1.3240659783724732e-06'	'1.3417485257710533e-06'	0	1
284	'1.4310950885672356e-07'	'2.2926680968197667e-07'	0	0
35	'1.3327634827936426e-06'	'1.2137290706280842e-06'	1	0
112	'2.6632623933910275e-07'	'3.3264805361940913e-07'	0	0
80	'1.7838337996946035e-06'	'1.8952782718480813e-06'	0	0
183	'1.238553066013824e-07'	'1.5489943899689191e-07'	0	0
28	'1.817280129288791e-06'	'1.1649931417562781e-06'	0	0
92	'2.583957548378655e-07'	'2.278586475944859e-07'	1	1
77	'8.894196101634929e-07'	'1.0630457185051212e-06'	0	1
222	'1.4972885365582447e-06'	'1.7449485013325266e-06'	0	0
81	'1.2243432985201187e-06'	'1.057473076306073e-06'	1	1
169	'1.3972702481749847e-06'	'1.4595228521238117e-06'	0	0
85	'5.837676431819273e-07'	'5.833072071924289e-07'	0	1
88	'9.298444418959419e-07'	'8.97358881177623e-07'	1	0
158	'1.9878768859349117e-06'	'1.9155249100822114e-06'	1	1
117	'8.09197422359686e-07'	'8.919853141162325e-07'	0	1
189	'1.1846441529277064e-06'	'8.871942043448793e-07'	1	1
129	'1.13701971174449e-06'	'1.2933083247859402e-06'	0	0
216	'2.666400915304204e-07'	'3.2265879361001367e-07'	0	0
168	'6.739242560788022e-07'	'8.476381784028494e-07'	0	0
94	'1.2946969608357322e-07'	'1.3153106768928557e-07'	0	1
19	'9.521475355852385e-07'	'9.90426150417701e-07'	0	0
11	'1.262028694672493e-06'	'9.696522921014406e-07'	1	0
261	'1.3615437699045387e-06'	'1.356413944300749e-06'	1	0
288	'1.04375761267731e-06'	'1.0575083955762023e-06'	0	0
56	'4.181304926013696e-07'	'4.98515883969838e-07'	0	0
103	'1.6759487037456792e-06'	'1.6918510831725192e-06'	0	1
293	'1.6245909819380920e-07'	'1.1804953304347585e-07'	1	1
138	'4.155810128378644e-07'	'4.92776146645009e-07'	0	0
68	'3.247372067263714e-07'	'1.0003420093384557e-07'	1	1

```
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Files
sample_data
Naive-Bayes-Classifica...
newTest.xlsx
newTestGroundTruth...
newTrain.xlsx
[50] {'ID': 111, 'Yes Probability': '1.9221871534145187e-06', 'No Probability': '2.812814122335686e-06', 'Prediction Result': 0, 'Ground Truth': 0}
{'ID': 108, 'Yes Probability': '7.68969458427846e-07', 'No Probability': '9.26758469328826e-07', 'Prediction Result': 0, 'Ground Truth': 1}
{'ID': 102, 'Yes Probability': '1.862825666720242e-06', 'No Probability': '1.7848899381730958e-06', 'Prediction Result': 1, 'Ground Truth': 0}
{'ID': 6, 'Yes Probability': '4.113636859824366e-07', 'No Probability': '2.8948916225434876e-07', 'Prediction Result': 1, 'Ground Truth': 1}
{'ID': 275, 'Yes Probability': '1.8178755714351752e-06', 'No Probability': '1.9759978466234326e-06', 'Prediction Result': 0, 'Ground Truth': 0}
{'ID': 227, 'Yes Probability': '1.0983345700710999e-07', 'No Probability': '1.5053198980780254e-07', 'Prediction Result': 0, 'Ground Truth': 0}
{'ID': 174, 'Yes Probability': '5.39178444286829e-07', 'No Probability': '5.395930588100931e-07', 'Prediction Result': 0, 'Ground Truth': 0}

TP : 18 FN : 20
TN : 29 FP : 22
Accuracy : 52.80898876404494%
Precision : 47.368421052631575%
Recall : 45.0%

[52] result = []

target = dataText # your ground truth data

result = doPrediction(yesMeanTrain, yesStdTrain, noMeanTrain, noStdTrain, target, columnTarget=['x1','x2','x3'], truthColumn='y')

for p in result:

    print(p)

confusionMatrix(result)
```

```
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Files
sample_data
Naive-Bayes-Classifica...
newTest.xlsx
newTestGroundTruth...
newTrain.xlsx
[52] {'ID': 297, 'Yes Probability': '2.8651683690601637e-07', 'No Probability': '4.2542525087080127e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 298, 'Yes Probability': '1.62777778734472986e-06', 'No Probability': '1.5669307209916147e-06', 'Prediction Result': 1, 'Ground Truth': '?'}
{'ID': 299, 'Yes Probability': '9.272157669291045e-07', 'No Probability': '1.053136177558649e-06', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 300, 'Yes Probability': '1.841980850095255e-07', 'No Probability': '4.6331689898113e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 301, 'Yes Probability': '2.8182801957802733e-07', 'No Probability': '3.877211056894782e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 302, 'Yes Probability': '2.028670588652634e-06', 'No Probability': '1.90148038914932e-06', 'Prediction Result': 1, 'Ground Truth': '?'}
{'ID': 303, 'Yes Probability': '1.8136232089548423e-07', 'No Probability': '2.040456260625972e-07', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 304, 'Yes Probability': '1.097694859373458e-06', 'No Probability': '1.3471838325743499e-06', 'Prediction Result': 0, 'Ground Truth': '?'}
{'ID': 305, 'Yes Probability': '1.8869489043817693e-07', 'No Probability': '1.7352676901958364e-07', 'Prediction Result': 1, 'Ground Truth': '?'}
{'ID': 306, 'Yes Probability': '1.457225156698698e-06', 'No Probability': '1.3016875296297746e-06', 'Prediction Result': 1, 'Ground Truth': '?'}

Cannot process the confusion matrix with unknown Ground Truth!

[55] exercisedata = pd.read_csv("Naive-Bayes-Classification-Data.csv")
exercisedata.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 995 entries, 0 to 994
Data columns (total 3 columns):
 #   column      Non-Null Count  Dtype
---  -
 0   glucose    995 non-null      int64
 1   bloodpressure  995 non-null      int64
 2   diabetes   995 non-null      int64
dtypes: int64(3)
memory usage: 23.4 KB

[56] exercisedata.describe()

      glucose  bloodpressure  diabetes
count  995.000000    995.000000    995.000000
```


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- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[56] exerciseData.describe()

	glucose	bloodpressure	diabetes
count	995.000000	995.000000	995.000000
mean	44.306533	79.184925	0.500503
std	6.707567	9.340204	0.500251
min	20.000000	50.000000	0.000000
25%	40.000000	72.000000	0.000000
50%	45.000000	80.000000	1.000000
75%	50.000000	87.000000	1.000000
max	70.000000	100.000000	1.000000

[57] exerciseData.head()

	glucose	bloodpressure	diabetes
0	40	85	0
1	40	92	0
2	45	83	1
3	45	80	0
4	40	73	1

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- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[58] id = [1 for i in range(len(exerciseData))]
exerciseData['id'] = id

[59] rescaledData = standardization(exerciseData, columnTarget=['glucose', 'bloodpressure'])
rescaledData.head()

	glucose	bloodpressure	diabetes	id
0	-0.642041	0.622585	0	0
1	-0.642041	1.372034	0	1
2	0.103386	-1.732823	1	2
3	0.103386	0.087265	0	3
4	-0.642041	-0.662183	1	4

[61] # Training
trainData, testData = folding(rescaledData, trainingPercentage=80, location='left', shuffle=True)
trainData

	glucose	bloodpressure	diabetes	id
0	0.103386	-0.662183	1	560
1	-0.642041	-0.233927	1	852
2	0.848813	-0.233927	1	535
3	0.848813	-1.839888	1	406

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- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[61]

	glucose	bloodpressure	diabetes	id
4	-0.642041	0.943778	0	494
...	...	...	...	...
791	-0.642041	0.622585	0	33
792	-0.642041	0.836714	0	330
793	-0.642041	0.622585	0	534
794	-0.642041	0.836714	0	827
795	-1.387468	0.408457	0	463

796 rows x 4 columns

[62] testData

	glucose	bloodpressure	diabetes	id
0	0.103386	0.622585	0	758
1	-0.642041	-2.054016	1	993
2	0.103386	-0.233927	1	519
3	0.103386	-0.126863	1	324
4	0.848813	0.408457	0	447
...	...	...	...	...
194	-1.387468	0.087265	1	14
195	-0.642041	-0.983375	1	273

196 rows x 4 columns

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- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[62]

	glucose	bloodpressure	diabetes	id
196	-0.642041	0.836714	0	71
197	-0.642041	1.479098	0	112
198	-0.642041	-0.448055	1	188

199 rows x 4 columns

[63] yesData, noData = splitTruth(trainData, columnTarget='diabetes')
yesData

	glucose	bloodpressure	diabetes	id
0	0.103386	-0.662183	1.0	560.0
1	-0.642041	-0.233927	1.0	852.0
2	0.848813	-0.233927	1.0	535.0
3	0.848813	-1.839888	1.0	406.0
5	1.594239	-1.197503	1.0	332.0
...	...	...	...	...
783	1.594239	-1.518695	1.0	512.0
784	-2.132894	-0.126863	1.0	80.0
785	-0.642041	0.622585	1.0	92.0
786	0.848813	0.087265	1.0	246.0
788	0.103386	-0.448055	1.0	166.0

406 rows x 4 columns

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Files

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- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

[64] noData

	glucose	bloodpressure	diabetes	id
4	-0.642041	0.943778	0.0	494.0
7	-0.642041	0.943778	0.0	445.0
8	0.103386	0.408457	0.0	320.0
9	1.594239	1.479098	0.0	766.0
12	0.103386	0.622585	0.0	197.0
...	...	...	...	...
791	-0.642041	0.622585	0.0	33.0
792	-0.642041	0.836714	0.0	330.0
793	-0.642041	0.622585	0.0	534.0
794	-0.642041	0.836714	0.0	827.0
795	-1.387468	0.408457	0.0	463.0

390 rows x 4 columns

[65] yesMean, noMean = find_mean(yesData, noData, columnTarget=['glucose', 'bloodpressure'])
yesMean
({'glucose': -0.0012676886494671466, 'bloodpressure': -0.7993093961188249})

[66] yesMean
({'glucose': -0.0012676886494671466, 'bloodpressure': -0.7993093961188249})

[67] yesStd, noStd = find_std(yesData, noData, columnTarget=['glucose', 'bloodpressure'])
yesStd
({'glucose': 1.3350011236792412, 'bloodpressure': 0.6612293457485487})

[68] noStd
({'glucose': 0.5641703550097849, 'bloodpressure': 0.4818315145149308})

[69] result = []
target = testData
result = doPrediction(yesMean, yesStd, noMean, noStd, target, columnTarget=['glucose', 'bloodpressure'], truthColumn='diabetes')
for p in result:

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Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
[69] for p in result:
      print(p)

confusionMatrix(result)

{'ID': 840, 'Yes Probability': '0.1199975720842184', 'No Probability': '2.2605310580927948e-05', 'Prediction Result': 1, 'Ground Truth': 1}
{'ID': 582, 'Yes Probability': '0.008420424788512806', 'No Probability': '0.5707305755007602', 'Prediction Result': 0, 'Ground Truth': 0}
{'ID': 39, 'Yes Probability': '0.005567439815218076', 'No Probability': '0.5477244805384819', 'Prediction Result': 0, 'Ground Truth': 0}
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Os completed at 9:33 PM

KERJA LAB NAIVE BAYES_130121 x +

colab.research.google.com/drive/1OUrEpApX9UVqbwjF0XcqypgP1lp137la#scrollTo=wwmEOYjgUJI

KERJA LAB NAIVE BAYES_1301213290.ipynb ☆

File Edit View Insert Runtime Tools Help All changes saved

Files

- sample_data
- Naive-Bayes-Classifica...
- newTest.xlsx
- newTestGroundTruth...
- newTrain.xlsx

Code

```
[70] for i in trainData.id:
      print(i)

746
359
369
525
334
956
96
922
144
652
759
388
279
184
190
281
825
924
345
950
4
```

Os completed at 9:33 PM

TP : 83 FN : 8
TN : 99 FN : 9
Accuracy : 91.4572864321608%
Precision : 91.20879120879121%
Recall : 90.21739130434783%

